Supplementary Material: Auditing Annotation Support and Evaluation Sensitivity in Aerial Vehicle Detection

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S1. Notation and Audit Quantities

This supplement specifies the formal objects and the data-free records underlying the manuscript. All reported quantities are restricted to the declared artifacts, mapping, common-coverage denominator, frozen candidate pool, and registered rule set. They are not estimates of annotation-error rates, a general detector leaderboard, or physical traffic density.

Let G_s be the valid mapped ground-truth boxes for source s, and let $G_s(r) \subseteq G_s$ be the boxes retained by a support rule r. The annotation-support shift is

$$S_s(r) = 1 - \frac{|G_s(r)|}{|G_s|}.$$

For candidate k and metric m, the fixed-(d, τ, c, ι) policy band is $B_m^\pi = \max_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota) - \min_{\pi \in \Pi_s} m(d, \tau, \pi, c, \iota)$. The authoritative primary policy set is $\Pi = \{I, O, N\}$: include-all (I), exclude/source-ignore-off (O), and exclude/source-ignore-on (N). The ancillary "exclude-and-ignore-removed" control is intentionally outside this set and therefore does not enter bands, headline record counts, or Fig. 2 of the main paper.

For a finite headline rule set $\mathcal{R}_h$, define the candidate-level rule band $B_k(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} m_k(r) - \min_{r \in \mathcal{R}_h} m_k(r)$. For a reference rule $r_0 \in \mathcal{R}_h$, winner $k^\star$, and competitor j, define $\Delta_j(r) = m_{k^\star}(r) - m_j(r)$ and

$$\Gamma_j(\mathcal{R}_h) = \max_{r \in \mathcal{R}_h} |\Delta_j(r) - \Delta_j(r_0)|.$$

$\Gamma_j(\mathcal{R}_h)$ is the differential policy radius. It removes common-mode score movement and is the relevant deterministic diagnostic for a pairwise rank claim. We additionally report $\min_{r \in \mathcal{R}_h} \Delta_j(r)$, the direct observed rank margin, and a paired-bootstrap winner probability for the declared resampling unit.

The main paper's rule tuple is $r = (d, \tau, \pi, c, \iota)$, where d selects absolute or normalized support, τ is the support threshold, π is a small-object policy, c is confidence, and ι is IoU. AP integrates over score order and hence has no fixed confidence coordinate. All multi-candidate comparisons are computed on C_s, defined before class filtering from raw-cache image names. A covered image with no retained vehicle prediction remains an evaluated empty prediction.

S2. Formal Results and Proofs

Lemma S1 (resize invariance). Under isotropic resizing by factor $\alpha > 0$, (w, h, W, H) maps to $(\alpha w, \alpha h, \alpha W, \alpha H)$. Thus $s'_a = \max(\alpha w, \alpha h) = \alpha s_a$, while

$$s'_n = \max\left(\frac{\alpha w}{\alpha W}, \frac{\alpha h}{\alpha H}\right) = s_n.$$

Consequently, a fixed-pixel threshold depends on image scale, whereas a normalized-side threshold is invariant to isotropic image resizing. This is an invariance statement about the audit quantity, not a claim that normalized filtering removes every source difference.

Proposition S1 (policy-robust candidate-pool winner). Suppose $k^\star$ is the reference winner and $\Delta_j(r_0) > 0$ for every competitor j. If $\Delta_j(r_0) > \Gamma_j(\mathcal{R}_h)$ for all j, then $k^\star$ is the winner for every $r \in \mathcal{R}_h$.

Proof. By the definition of $\Gamma_j(\mathcal{R}_h)$,

$$\Delta_j(r) \geq \Delta_j(r_0) - |\Delta_j(r) - \Delta_j(r_0)| \geq \Delta_j(r_0) - \Gamma_j(\mathcal{R}_h) > 0.$$

Thus $m_{k^\star}(r) > m_j(r)$ for every competitor and every registered rule. □

Corollary S1. Since

$$\Gamma_j(\mathcal{R}_h) \leq B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h),$$

the sufficient condition $\Delta_j(r_0) > B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$ guarantees the same conclusion.

Proof. Write $x_k(r) = m_k(r) - m_k(r_0)$. Each $x_k(r)$ lies between the minimum and maximum movement of candidate k, whose range over $\mathcal{R}_h$ is $B_k(\mathcal{R}_h)$. Hence $|x_{k^\star}(r) - x_j(r)| \leq B_{k^\star}(\mathcal{R}_h) + B_j(\mathcal{R}_h)$, and maximization over $r \in \mathcal{R}_h$ gives the inequality. □

The corollary is deliberately conservative: it ignores correlation in rule-induced movements. For example, let $(m_A(r_0), m_B(r_0)) = (.70, .68)$ and, under another rule, $(.60, .58)$. Both individual bands equal .10, exceeding the

TABLE S1
Notation used in the supplementary analysis.

Symbol	Definition
$G_s, G_s(r)$	Valid mapped boxes; boxes retained by rule r
$S_s(r)$	Annotation-support shift $1 - \lvert G_s(r)\rvert / \lvert G_s\rvert$
$m_k(r), B_m^\pi$	Candidate metric; fixed-condition policy band
$B_k(\mathcal{R}_h)$	Candidate rule band over the headline rule set
$\Delta_j(r)$	Winner–competitor margin under rule r
$\Gamma_j(\mathcal{R}_h)$	Maximum margin change over $\mathcal{R}_h$ from r_0
C_s	Source-wise common image-coverage denominator
T, P, N	Global TP, FP, and FN counts for micro-F_1
D	$2T + P + N$, the micro-F_1 denominator

TABLE S2
PAIRWISE DIAGNOSTICS OVER SIX NOMINAL SCALE–POLICY SETTINGS. EVERY ROW HAS A DETERMINISTIC CERTIFICATE; BOOTSTRAP QUALIFICATION IS REPORTED SEPARATELY.

Source	Metric	Δ_0	Γ	$\min\Delta$	Cert.	Bootstrap
VisDrone	AP_{50}	.3544	.0206	.3544	yes	all > 0
Local UAVDT	AP_{50}	.0340	.0326	.0014	yes	all include 0
VisDrone	F_1	.3354	.0026	.3354	yes	all > 0
Local UAVDT	F_1	.1911	.00007	.1910	yes	all include 0

reference margin .02, yet $\Delta_B(r) = .02$ throughout and $\Gamma_B = 0$. Conversely, anti-aligned movements can make small individual bands consequential for a close race. Differential radii therefore complement, rather than replace, the individual policy bands.

S3. Coverage Identifiability and an F_1 Bound

Proposition S2 (non-identifiability of full-population micro-F_1 ranking under unequal coverage). Let candidates A and B be observed on different image subsets C_A and C_B. If their TP, FP, and FN contributions on unobserved images are unconstrained, the ordering of their full-population micro-F_1 scores is not identifiable from the observed subset counts.

Proof. Keep all observed counts fixed. On unobserved images, construct one completion with sufficiently favorable TP/FP/FN contributions for A and unfavorable contributions for B; it produces an A-leading full-population micro-F_1. Reverse these unconstrained contributions to produce a B-leading completion with exactly the same observed subset counts. Thus no unique full-population micro-F_1 ordering is identified. □

An analogous construction applies to AP when scores and detections on unobserved images are unrestricted; it is not needed for the formal micro-F_1 result above.

This result motivates the source-wise common denominator used for VisDrone and local UAVDT. It also gives the formal boundary for AI-TOD: its 1,869 covered images have 16,900 vehicle boxes (9.0 per image; median side 12 px), while 935 noncovered images have 43,004 boxes (46.0 per image; median side 16 px). The observed coverage-conditioned sensitivity analysis cannot be extrapolated into a full-source ranking without missing-coverage assumptions.

Proposition S3 (count-perturbation bound for micro-F_1). For $F(T, P, N) = 2T/(2T + P + N)$, let $D_{\min} > 0$ lower-bound $D = 2T + P + N$ along the segment between two nonnegative count vectors. Then

$$|\Delta F| \leq \frac{2|\Delta T| + |\Delta P| + |\Delta N|}{D_{\min}}.$$

Proof. The partial derivatives satisfy $|\partial F/\partial T| = 2(P + N)/D^2 \leq 2/D$, and $|\partial F/\partial P| = |\partial F/\partial N| = 2T/D^2 \leq 1/D$. Apply the integral form of the mean-value bound along the line segment and use $D \geq D_{\min}$. □

The bound does not predict an observed band from support removal alone, because the signs and magnitudes of TP, FP, and FN changes remain detector- and policy-dependent. It does explain why extreme removal permits a large band: at 24 px, the pooled excluded fractions are 26.7%, 4.9%, and 90.3% for VisDrone, local UAVDT, and AI-TOD, respectively. AI-TOD's .330 F_1 band is therefore interpreted jointly with retained support and removed-object treatment, not as an isolated detector effect.

TABLE S3
ARTIFACT-LEVEL PROVENANCE AND EVALUATION SCOPE.

Source	Eval. imgs	Covered	Vehicle boxes	Mapping	Scope
VisDrone	548	548	17,040	car/van/truck/bus	full audit
Local UAVDT	305	305	11,054	car/truck/bus	full audit
AI-TOD	2,804	1,869	59,904	vehicle	sensitivity only
DOTA-v2	5,108	–	53,349	small/large vehicle	structural only

S4. Artifact Provenance and Coverage Audit

Table S3 records the source-specific boundaries that matter for interpretation. DOTA-v2 contributes only structural support: its quadrilaterals are converted to axis-aligned horizontal-box proxies for the scale audit, so its distances are representation-conditioned. The local UAVDT artifact is a COCO-style conversion and is conversion-conditioned. AI-TOD's available crop reconstruction is xView-derived and has incomplete prediction coverage; it is not used for a rank claim.

The vehicle-positive image counts are 519/548 for VisDrone, 305/305 for local UAVDT, 1,443/2,804 for AI-TOD, and 2,581/5,108 for DOTA-v2. Median horizontal-box maximum sides are 41, 33, 14, and 18 px, respectively; median normalized sides are .0412, .0278, .0175, and .0352. These are descriptive support properties. They do not establish which source is "more correct," nor do they assert a causal resolution effect.

Raw prediction coverage is determined by image-name presence before mapping predictions into the vehicle superclass. Consequently, a covered image whose predictions are entirely filtered out still contributes all valid mapped ground truth as false negatives, rather than vanishing from AP or F_1. Source ignores are preserved for VisDrone. Its truncation and occlusion attributes are not filtered. For local UAVDT, the applied coordinates are the recorded $xywh$ conversion. The provenance manifest in the repository supplies raw paths and hashes without redistributing licensed images or prediction caches.

The published package at https://anonymous.4open.science/r/aerial-evaluation-audit-5565/README.md contains the ontology mapping, candidate manifest, coverage records, corrected headline outputs, matching controls, bootstrap records, figure-source summaries, scripts, and SHA-256 manifest. It is sufficient to inspect every claim-scoped derived record; acquiring the licensed source artifacts remains the responsibility of the reproducer.

S5. Full Rule Contract and Matching Controls

The full greedy surface comprises seven candidates, eight confidence values, four IoUs, eight absolute or six normalized support thresholds, and three primary policies: $7[8 \cdot 4 \cdot 8 \cdot 3 + 8 \cdot 4 \cdot 6 \cdot 3] = 9{,}408$ records. The absolute thresholds are $\{16, 20, 24, 28, 32, 40, 48, 64\}$ px, normalized thresholds are $\{.005, .0075, .010, .015, .020, .030\}$, confidence values are $\{.10, .15, .20, .25, .30, .35, .40, .50\}$, and IoUs are $\{.25, .30, .40, .50\}$. Headline rules use 24 px and .015, registered interior grid values selected before headline computation.

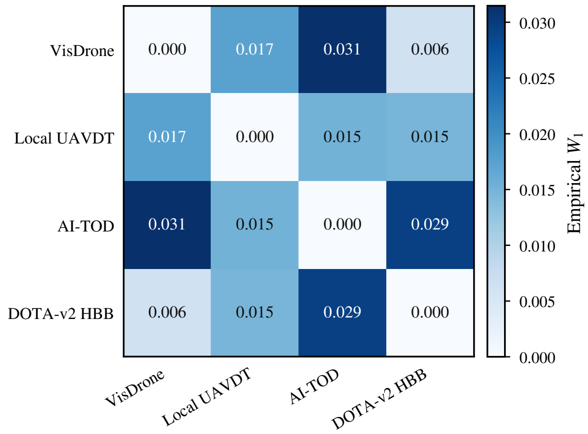

Fig. S1. Equal-box-weight empirical 1-Wasserstein distances on normalized horizontal-box maximum side. DOTA-v2 uses an axis-aligned proxy derived from quadrilaterals; its distances are representation-conditioned. The matrix is a structural diagnostic rather than a performance result.

TABLE S4
POLICY SEMANTICS AND ROLE IN REPORTED QUANTITIES.

Policy	Removed valid boxes	Use
Include-all	retained	primary grid
Exclude/source-ignore-off	absent; overlaps may be FP	primary grid
Exclude/source-ignore-on	absent; source ignores suppress overlaps	primary grid
Exclude-and-ignore-removed	absent; removed overlaps suppressed	ancillary diagnostic

For fixed-operation F_1, predictions are score ordered. Each prediction matches the highest-IoU eligible yet-unmatched valid ground-truth box above the declared IoU; unmatched predictions are FP and unmatched retained boxes are FN. The Hungarian control first removes invalid edges, maximizes cardinality of valid matches, and then maximizes total IoU among maximum-cardinality solutions. Dummy nodes permit unmatched detections and boxes. Maximizing cardinality before total IoU prevents an IoU-optimal assignment from reducing the number of valid matches.

AP uses one-category COCO bbox evaluation with complete score order, IoU .5 for AP_{50}, standard area ranges, and registered `maxDets` $= 100$. The 100-detection cap is material: after source-ignore handling, it truncates predictions in 406–490 VisDrone images and 272–292 local-UAVDT images across primary policies. Focused 300 and 2000 controls preserve every one of the 10 nonduplicate multi-candidate AP winners; at 2000 no retained prediction is truncated (the largest retained count is 1,066), but AP_{50} pairwise gaps relative to 100 shift by $[-.041, .048]$.

The ancillary control is especially informative for AI-TOD: under it, F_1 is .130 at 24 px and .317 at .015, compared with .065 and .260 under exclusion without protection. Thus the observed policy variation combines extreme support removal and the treatment of predictions that overlap removed targets.

S6. EVALUATION CONTROLS AND BOUNDARY CONDITIONS

The main paper reports the complete F_1 policy-band surfaces and the paired uncertainty geometry. Figure S2 collects three boundary-condition checks derived from the same released records; it introduces neither new candidates nor new rules.

The matcher comparison shows that winner agreement does not imply metric identity. The detection-cap panel likewise preserves the pairwise winner while moving the gap: relative to `maxDets`=100, gaps at 2000 shift by $-.041$ for VisDrone and $+.048$ for local UAVDT under the displayed reference rule. Panel (c) shows why no AI-TOD rank or full-source uncertainty statement is made: the 935 noncovered images contain more boxes and a markedly higher boxes-per-image rate than the covered subset.

S7. COMPLETE HEADLINE BOOTSTRAP RECORDS

Tables S5–S6 make the uncertainty qualification explicit. Each row is a separate paired bootstrap interval; endpoint ranges in the main paper are summaries across these rows, not a single pooled confidence interval. The local-UAVDT rows use the filename-derived sequence proxy described in the main paper and Section S8.

The apparent duplication in some source-ignore rows is itself an audit finding. Source-ignore-on and source-ignore-off are identical for local UAVDT under the registered artifacts, so four nominal AP outcomes collapse to fewer distinct policy semantics. We retain nominal row-level records for transparent replay and separately state the 10 nonduplicate multi-candidate AP outcomes and 20 nonduplicate matching outcomes in the main paper.

S8. BOOTSTRAP, REPRODUCIBILITY, AND CLAIM AUDIT

Pairwise uncertainty resamples the common evaluation unit jointly for the leading and runner-up candidates. VisDrone uses image units. Local UAVDT uses a filename-derived sequence proxy (substring before the first underscore): 93 units over 305 images, with min/median/max unit sizes of 1/1/104. Each analysis uses 1,000 paired replicates, seed 20260712, and percentile 2.5/97.5% intervals. Replicates recompute global micro-F_1 from resampled TP/FP/FN totals and recompute score-ordered one-category COCO AP_{50} at `maxDets` $= 100$.

For AP_{50}, all six VisDrone policy-specific intervals favor SAHI: their lower endpoints range from .339 to .362 and their upper endpoints from .366 to .387. Every one of the four nonduplicate local-UAVDT sequence-proxy intervals contains zero, with lower endpoints from $-.310$ to $-.296$, upper endpoints .071–.111, and winner probabilities .471–.604. For F_1, all 12 VisDrone intervals are positive with winner probability 1.00, whereas all 12 local-UAVDT sequence-proxy intervals contain zero despite deterministic tiling wins (winner probabilities .691–.811). The latter unit is a sensitivity device, not evidence that filename prefixes recover official UAVDT sequence identities.

Reproducibility and claim-audit contract.

Item	Released or fixed record
Rule contract	`configs/locked_rule_grid.yaml`
Ontology/mapping	`label_reliability/config/ontology.json`
Coverage denominator	`outputs/coverage/*.csv`
AP/F_1 headline records	`outputs/ap_headline`, `coverage_corrected_grid`
Matcher control	`outputs/matching_headline/*.csv`
Bootstrap outputs	`outputs/statistics/bootstrap_headline_*.csv`
Policy diagnostics	`pairwise_policy_robustness.csv`
Integrity check	`SHA256SUMS`; claim audit JSON

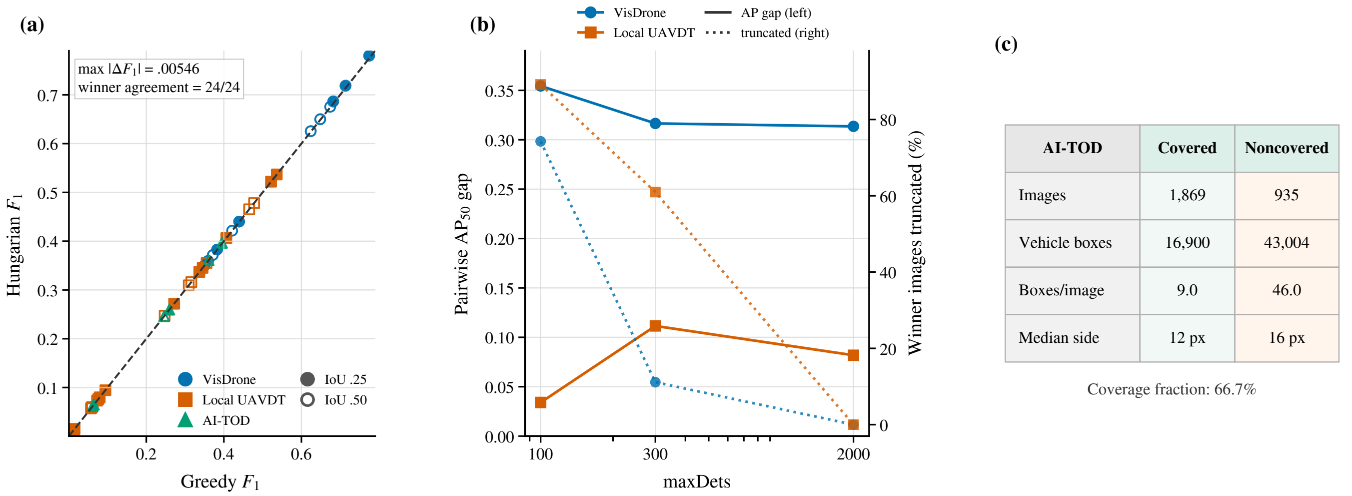

Fig. S2. Evaluation controls and boundary conditions. (a) Greedy versus threshold-first Hungarian F_1 at confidence .25; marker shape denotes source and fill denotes IoU. The 24 multi-candidate checks have identical winners although metric values are not identical. (b) At the include-all absolute-24-px reference rule, solid lines show the VisDrone SAHI–baseline and local-UAVDT tiling–baseline AP_{50} gaps; dotted lines show the fraction of images truncated for the corresponding winning candidate. (c) AI-TOD covered versus noncovered support, demonstrating that its single-candidate analysis is coverage-conditioned.

TABLE S5

AP_{50} PAIRED-BOOTSTRAP HEADLINE RECORDS. DIFFERENCES ARE WINNER MINUS RUNNER-UP.

Source	Rule	Threshold	Policy	Winner	Difference	95% interval	Winner p
VisDrone	absolute	24 px	include-all	SAHI	.354	[.339,.366]	1.000
VisDrone	absolute	24 px	exclude/off	SAHI	.373	[.361,.385]	1.000
VisDrone	absolute	24 px	exclude/on	SAHI	.373	[.361,.385]	1.000
VisDrone	normalized	.015	include-all	SAHI	.354	[.339,.366]	1.000
VisDrone	normalized	.015	exclude/off	SAHI	.375	[.363,.386]	1.000
VisDrone	normalized	.015	exclude/on	SAHI	.375	[.362,.387]	1.000
Local UAVDT	absolute	24 px	include-all	tiling	.034	[-.296,.108]	.585
Local UAVDT	absolute	24 px	exclude/off	tiling	.028	[-.310,.102]	.577
Local UAVDT	normalized	.015	include-all	tiling	.034	[-.303,.111]	.604
Local UAVDT	normalized	.015	exclude/off	tiling	.001	[-.303,.071]	.471

Greedy matching pseudocode. Sort retained predictions by descending score. For each prediction, form eligible unmatched GT boxes with IoU at least ι; if the set is nonempty, match its highest-IoU member and increment TP, otherwise increment FP. After all predictions, unmatched retained GT boxes increment FN. Input order is never used to resolve a metric tie: exact candidate ties are recorded as rank instability.

Threshold-first Hungarian pseudocode. Build the prediction–GT bipartite graph only from edges with IoU at least ι. First maximize cardinality; among maximum-cardinality assignments, maximize summed IoU; use dummy nodes for unmatched objects. Count matched valid edges as TP, unmatched predictions as FP, and unmatched GT boxes as FN. The released test suite contains a counterexample for cardinality-first optimization, plus no-op policy, input-order, and prediction-order checks.

Claim group	Derived artifact	Records	Validation
Coverage/support	coverage and scale audits	4 sources	exact counts/hashes
AP/F_1 policy results	headline tables and grids	42/9,408	numerical audit
Matching control	Hungarian agreement file	84	winner/tie checks
Uncertainty	paired bootstrap tables	34	seed and interval checks
Detection cap	maxDets controls	20	truncation/gap checks

Detection-cap audit. The released truncation table records, for every multi-candidate AP setting, the number of images affected and detections removed at `maxDets` = 100, 300, and 2000. At 2000, all ten retained-prediction settings have zero truncation; the largest per-image retained prediction count is 1,066. The paired AP_{50} winner is unchanged in all ten nonduplicate outcomes, but the corresponding winner–runner-up gaps move by $[-.041, .048]$ from 100 to 2000. This control is reported as evaluator sensitivity, not as a recommendation to replace the registered 100-detection COCO setting.

Claim-audit summary. The 55 verified numerical claims comprise artifact counts and support summaries (16), AP and F_1 policy quantities (18), matching and tie-sensitive quantities (5), bootstrap endpoints/probabilities (10), and detection-cap controls (6). The audit checks the manuscript's rounded values

TABLE S6
COMPLETE MICRO-F_1 PAIRED-BOOTSTRAP HEADLINE RECORDS. DIFFERENCES ARE WINNER MINUS RUNNER-UP; EACH ROW USES 1,000 PAIRED REPLICATES.

Source	Rule	Threshold	Policy	IoU	Winner	Difference	95% interval	Winner p
VisDrone	absolute	24 px	include-all	0.25	SAHI	0.335	[0.328,0.342]	1.000
VisDrone	absolute	24 px	include-all	0.50	SAHI	0.253	[0.241,0.264]	1.000
VisDrone	absolute	24 px	exclude/off	0.25	SAHI	0.322	[0.316,0.330]	1.000
VisDrone	absolute	24 px	exclude/off	0.50	SAHI	0.274	[0.265,0.284]	1.000
VisDrone	absolute	24 px	exclude/on	0.25	SAHI	0.323	[0.316,0.330]	1.000
VisDrone	absolute	24 px	exclude/on	0.50	SAHI	0.275	[0.265,0.284]	1.000
VisDrone	normalized	.015	include-all	0.25	SAHI	0.335	[0.328,0.342]	1.000
VisDrone	normalized	.015	include-all	0.50	SAHI	0.253	[0.241,0.264]	1.000
VisDrone	normalized	.015	exclude/off	0.25	SAHI	0.331	[0.324,0.338]	1.000
VisDrone	normalized	.015	exclude/off	0.50	SAHI	0.277	[0.267,0.287]	1.000
VisDrone	normalized	.015	exclude/on	0.25	SAHI	0.332	[0.325,0.339]	1.000
VisDrone	normalized	.015	exclude/on	0.50	SAHI	0.277	[0.268,0.286]	1.000
Local UAVDT	absolute	24 px	include-all	0.25	tiling	0.191	[-0.238,0.282]	0.786
Local UAVDT	absolute	24 px	include-all	0.50	tiling	0.162	[-0.258,0.259]	0.735
Local UAVDT	absolute	24 px	exclude/off	0.25	tiling	0.185	[-0.229,0.275]	0.811
Local UAVDT	absolute	24 px	exclude/off	0.50	tiling	0.156	[-0.251,0.251]	0.716
Local UAVDT	absolute	24 px	exclude/on	0.25	tiling	0.185	[-0.243,0.275]	0.749
Local UAVDT	absolute	24 px	exclude/on	0.50	tiling	0.156	[-0.261,0.257]	0.725
Local UAVDT	normalized	.015	include-all	0.25	tiling	0.191	[-0.238,0.281]	0.808
Local UAVDT	normalized	.015	include-all	0.50	tiling	0.162	[-0.250,0.256]	0.721
Local UAVDT	normalized	.015	exclude/off	0.25	tiling	0.135	[-0.231,0.218]	0.775
Local UAVDT	normalized	.015	exclude/off	0.50	tiling	0.108	[-0.256,0.196]	0.691
Local UAVDT	normalized	.015	exclude/on	0.25	tiling	0.135	[-0.234,0.217]	0.782
Local UAVDT	normalized	.015	exclude/on	0.50	tiling	0.108	[-0.235,0.194]	0.705

against named released derived artifacts; it does not certify the external licensed inputs. Each candidate manifest entry records a path, inclusion decision, and SHA-256 value, while `SHA256SUMS` validates the data-free supplementary repository files.

A minimal reproduction order is: verify hashes; inspect coverage and the frozen candidate manifest; evaluate AP and coverage-corrected absolute/normalized grids; run threshold-first Hungarian and removed-as-ignore controls; run paired bootstraps; regenerate the pairwise-policy diagnostic; and compile the manuscript. The data-free supplementary repository deliberately omits licensed raw imagery and prediction Parquet files while retaining their manifest paths and hashes. The automated manuscript claim audit verifies 55 numerical claims against released derived records. This makes the scope of each figure, table, and headline statement inspectable without extending the declared candidate pool or rerunning model training.

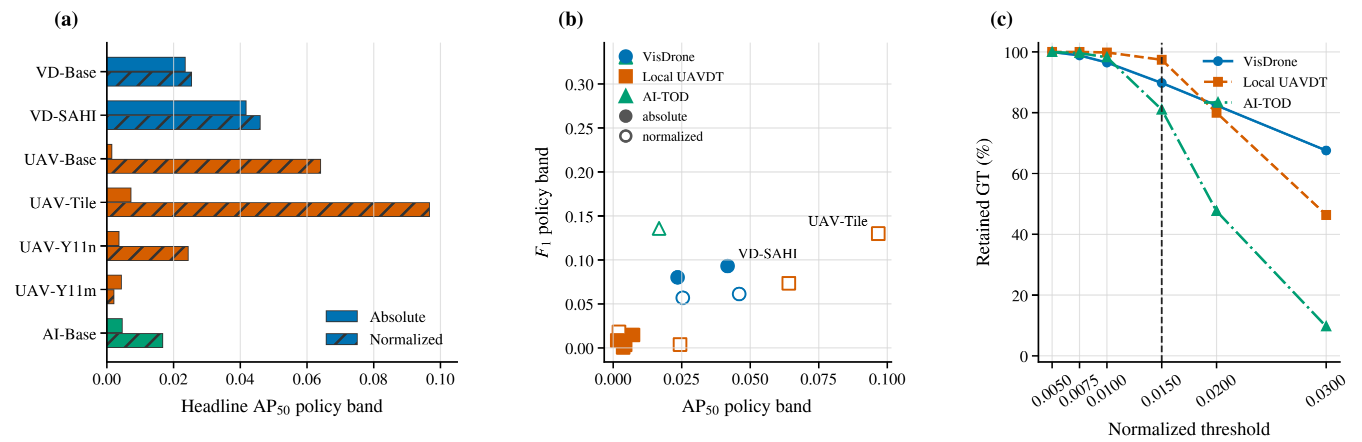

Fig. S3. Additional headline AP_{50} and support diagnostics. (a) Policy bands for all seven frozen candidates at the absolute-24-px and normalized-.015 rules; color denotes source and hatching denotes normalized support. (b) Headline AP_{50} versus F_1 policy bands at confidence/IoU .25 for F_1; source is encoded by marker shape/color and normalized rules are hollow. Labels identify the two multi-candidate pool winners. (c) Empirical retained-GT curves over the complete normalized-threshold grid. These are headline AP bands, not an uncomputed full AP surface.

S9. Additional AP and Support Diagnostics

Figure S3 separates two complementary effects. The normalized headline rule enlarges the local-UAVDT AP_{50} band from .007 to .097 for tiling and from .001 to .064 for the baseline. By comparison, the VisDrone SAHI bands are .042 and .046 under absolute and normalized support, and the single AI-TOD candidate changes from .005 to .017. The plot therefore reports candidate-specific evaluator sensitivity; it does not compare AP levels across sources.

Panel (c) shows the target-set movement behind those differences. At the headline .015 threshold, the retained fractions are 89.8%, 97.4%, and 81.0% for VisDrone, local UAVDT, and AI-TOD. At .030 they separate more strongly, so normalized support is resize-invariant without being source-equalizing. A full AP threshold surface was not materialized and is not claimed; the released AP evidence is restricted to the two headline support rules, while the complete threshold–confidence sensitivity surface in the main paper is based on F_1.

Exact headline policy bands. The table below gives absolute metric differences, not percentages. AP_{50} uses complete score ordering at `maxDets=100`; F_1 uses confidence/IoU .25.

Candidate	AP_{50} abs.	AP_{50} norm.	F_1 abs.	F_1 norm.
VD-Base	.02350	.02539	.08020	.05695
VD-SAHI	.04173	.04597	.09308	.06130
UAV-Base	.00149	.06407	.00852	.07347
UAV-Tile	.00722	.09671	.01474	.12989
UAV-Y11n	.00363	.02439	.00058	.00383
UAV-Y11m	.00436	.00211	.00356	.01821
AI-Base	.00462	.01674	.33033	.13588

Complete empirical retention grid. Percentages below are pooled box-level fractions computed before any prediction matching. They expose the support target associated with every registered scale threshold and can be reproduced directly from `box_scale_records.parquet`.

Rule	Threshold	VisDrone	UAVDT	AI-TOD
Abs.	16 px	85.7	99.6	47.6
Abs.	20 px	79.5	98.8	20.4
Abs.	24 px	73.3	95.1	9.7
Abs.	28 px	67.5	77.5	5.8
Abs.	32 px	62.1	56.4	4.2
Abs.	40 px	52.3	34.3	2.5
Abs.	48 px	42.7	30.0	1.4
Abs.	64 px	27.7	17.7	0.4
Norm.	.0050	100.0	100.0	99.9
Norm.	.0075	98.8	100.0	99.6
Norm.	.0100	96.5	99.8	98.2
Norm.	.0150	89.8	97.4	81.0
Norm.	.0200	82.4	80.0	47.6
Norm.	.0300	67.6	46.4	9.7